

6.7 Investments



Figure 6.13 Stocks are bought and sold to improve investment values. (credit: modification of work "FT ringing the Closing Bell at the NYSE" by Financial Times/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Distinguish between basic forms of investments including stocks, bonds, and mutual funds.
2. Understand what bonds are and how bond investments work.
3. Understand how stocks are purchased and gain or lose value.
4. Read and derive information from a stock table.
5. Define a mutual fund and how to invest.
6. Compute return on investment for basic forms of investments.
7. Compute future value of investments.
8. Compute payment to reach a financial goal.
9. Identify and distinguish between retirement savings accounts.

You can save your money in a safe or a vault (or worse, under the mattress!), but that money does not grow. It would be hard to save enough for retirement that way. What can be done to increase the value of the money you already have?

The answer is to invest it. Use the money that you have to earn more money back. For instance, as we saw in [Methods of Savings](#), you can save it in a bank. Or, to reach loftier goals, invest in something more likely to grow, such as stocks.

A great example of this is Apple stock. Anyone who bought stock in Apple Inc. (formerly Apple Computer, Inc.) in 1997 and held onto the shares earned a lot of money. [To be more specific, \\$100 worth of Apple shares bought in 1980, when it was first sold to the public, was valued at \\$67,564 in 2019, or 676 times more! \(\[https://openstax.org/r/much_aspx\]\(https://openstax.org/r/much_aspx\)\)](#)

Perhaps you have heard a story like that, of an investment opportunity taken that paid off, or the story of an investment opportunity missed. But such stories are the exceptions.

In this section, we'll investigate bonds, stocks, and mutual funds and their comparative strengths and weaknesses. We close the section with a discussion of retirement savings accounts.

Distinguish Between Basic Forms of Investments

Bonds, stocks, and mutual funds tend to offer higher returns, but to varying degrees, come with higher risks. Stocks and mutual funds also vary in how much they earn. Their predicted rates of return on investment are not guaranteed, but educated guesses based on market trends and historical performance.

We will use the methods and formulas we learned earlier to evaluate these forms of investment.

Bonds

Bonds are issued from big companies and from governments. Selling bonds is an alternative to an institution taking a loan from a bank. The funds from the selling of bonds are often used for large projects, like funding the building of a new highway or hospital.

Bonds are considered a conservative investment. They are bought for what is known as the **issue price**. The interest is fixed (does not change) at the time of purchase and is based on the issue price of the bond. The interest rate is often referred to as the coupon rate; the interest paid is often called the coupon yield. The interest paid is often higher than savings accounts and the risk is exceptionally low. The bond is for a fixed length of time. The end of this time is the

maturity date of the bond.

There are several types of bonds:

- **Treasury bonds** are issued by the federal government.
- **Municipal bonds** are issued by state and local governments.
- **Corporate bonds** are issued by major corporations.

There are other types of bonds available, but they are beyond the scope of this section.

?? WHO KNEW?

Trading Bonds

Bonds are often part of larger investment portfolios. These bonds may be traded. However, the interest paid is based on the price when the bond was bought (the issue price). These bonds can be bought and sold for more or less money than the issue price. If the bond is bought for more than the issue price, the interest is still paid on the issue price, not on the purchase price when the trade was made. This means the actual return on the bond decreases. If the bond is bought for less than the issue price, the return on the bond goes up.

▶ VIDEO

[Bonds \(https://openstax.org/r/investing_basics_bonds\)](https://openstax.org/r/investing_basics_bonds)

EXAMPLE 6.62

Bond Investment

Muriel purchases a \$3,000 bond with a maturity of 4 years at a fixed coupon rate of 5.5% paid annually. How much is Muriel paid each year, and how much does she receive on the maturity date?

✓ Solution

The coupon rate is 5.5%. 5.5% of her bond value is $0.055 \times \$3,000 = \165 . After year 1, Muriel receives \$165. She receives \$165 after years 2 and 3 also. In year 4, when the bond matures, Muriel receives \$3,165, or the interest and the initial investment, or principal.

> YOUR TURN 6.62

1. Maureen invests \$5,000 in a bond with a maturity date in 5 years at a fixed coupon rate of 4.75%. How much is Maureen paid each year and how much does she receive on the maturity date?

Stocks

Stocks are part ownership in a company. They come in units called **shares**. The performance and earnings of stocks is not guaranteed, which makes them riskier than any other investment discussed earlier. However, they can offer higher return on investment than the other investments. Their value grows in two ways. They offer **dividends**, which is a portion of the profit made by the company. And the price per share can increase based on how others see that value of the company changing. If the value of the company drops, or the company folds, the money invested in the stock also drops.

Most stock transactions are executed through a broker. Brokers' commissions can be a percentage of value of the trades made or a flat fee. There are full-service brokers who charge higher commission rates, but they also offer financial advice and perform the research that you may not have the time or the expertise to do on your own. A discount broker only executes the stock transactions, buying or selling, so they charge lower rates than full-service brokers. There are also brokers that offer commission-free trading.

An important thing to remember is that stocks might provide a very large return on investment, but the trade-off is the risk associated with owning stocks.

? WHO KNEW?**Chapter 11 Bankruptcy and Stocks**

In the fall of 2022, the parent company of Regal Theaters, named Cineworld, filed for Chapter 11 bankruptcy. According to news articles, the bankruptcy was necessitated due to its heavy debt load. Generally, a company can file for Chapter 11 bankruptcy to allow them time to reorganize and restructure debts. When this happens, the company, after the Chapter 11 process is over, offers new stock. This makes the previous stock worthless. However, the company may allow an exchange of old stock for a discounted amount of the new stock. This in effect reduces (maybe vastly) the wealth held by those who owned the original stock.

EXAMPLE 6.63**Buying Stock in Company ABC**

Haniah buys stock in the ABC company, investing a total of \$13,000. She expects the stock to grow, through stock price increase and reinvestment of dividends, by 12.3% per year and compounded annually. If she leaves that money invested, how much will the stocks be worth in 20 years?

✓ Solution

Calculating this is a compound interest calculation, if Haniah's assumption about the stock's performance is correct. If so, then the principal is \$13,000, the rate is 0.123, the number of compounding periods per year is 1, and the time is 20 years. Substituting into the compound interest formula from [Methods of Savings](#), and computing, we have

$A = P\left(1 + \frac{r}{n}\right)^{nt} = \$13,000\left(1 + \frac{0.123}{1}\right)^{1 \times 20} = \$13,000 \times 10.1764223996 = \$132,293.49$. After 20 years, her stock is now worth \$132,293.49.

> YOUR TURN 6.63

1. Rixie deposits \$23,000 in the stock of DEF company. She assumes the stock value to grow though stock price increase and reinvestment of dividends by 13.8% compounded annually. How much will her stock be worth in 12 years?

? WHO KNEW?**Risk and Volkswagen**

The question of risk hovers over every investment. How risky can it get? Volkswagen seems to be a rather safe investment. But in 2015, Volkswagen's stock tumbled 30% over a few days when it was revealed that the company had installed software that altered the emission performance of some of their diesel engines. Volkswagen's hope was that lower emissions would bolster US sales of some of their diesel models. This was a drastic drop, and many investors lost a lot of money. However, the stock has come back since then. This was mild compared to the 65% drop in the Martha Stewart Living Omnimedia stocks.

**PEOPLE IN MATHEMATICS****Warren Buffett**

Warren Buffett is an investment legend. He began his career as an investment salesman in the 1950s. He formed Buffett associates in 1956. In 1965, he was in control of Berkshire Hathaway, which began as a merger between two textile companies. In his role there, he began to invest in a variety of companies. It is now a conglomerate holding company, and fully owns GEICO, Duracell, Dairy Queen, and other large companies.

His investment philosophy involves finding stocks and bonds from companies that have high intrinsic worth compared to their stock or bond prices. This means he focuses not on the supply and demand side of stock investing, but instead on the company's worth in total. Using this philosophy, he has become one of the world's most successful investors.

Reading Stock Tables

Information about particular stocks is contained in **stock tables**. This information includes how much the stock is selling for, and its high and low values from the past year (52 weeks). In a newspaper, the stock table may look like this:

52-Week High Low		Stock	SYM	Div	Yld %	P/E	Vol 100s	High	Low	Close	Net Chg
41.66	18.90	McDonald's	MCD	.72	2.9	12	7588	25.73	23.87	25.42	+0.31
22.60	13.20	Monsanto	MON	.52	2.4	55	15474	21.86	21.48	21.64	-0.29
17.05	8.30	Motorola	MOT	.16	1.7	dd	16149	10.57	8.88	10.43	+0.14
31.75	22.99	Mueller	MLI	-	-	16	1564	29.32	27.03	27.11	-0.02

Table 6.2 excerpt from a stock table, 2008

The symbols and abbreviations are defined here:

52-week High	52-week Low	The highest and lowest price of the stock over the past 52 weeks
Stock	SYM	The name of the company and the symbol used for trading
	Annual DIV	The current annual dividend per share
	Yld %	Percent yield is = $\frac{\text{annual dividend}}{\text{share price}} \times 100$
	P/E	Price to earnings ratio, share price divided by earnings per share over past year (dd indicates loss)
	Vol 100s	The number of shares traded yesterday in 100s
High	Low	The highest and lowest prices at which stocks traded yesterday
	Close	The price at which the stock traded at the close of the market yesterday
	Net Chg	Net change; change in price from market close 2 days ago to yesterday's close

The formulas for yield and price to earnings is a good way to measure how much the stock returns per share. Their values are calculated in the stock table, but deserve attention here.

FORMULA

The price to earnings ratio of a stock, P/E, is $P/E = \frac{\text{Share Price}}{\text{Dividend}}$. The percent yield for a stock, Yld%, is $Yld\% = \frac{\text{Annual Dividend}}{\text{Share Price}} \times 100\%$.

It should be noted that the price of a stock increases and decreases every moment, and so these value change as the share price changes.

EXAMPLE 6.64

Computing Percent Yield

1. Find the percent yield for a stock with a price of \$30.69 and an annual dividend of \$1.48.
2. Find the percent yield for a stock with a price of \$62.25 and an annual dividend of \$1.76.

Solution

1. Substituting the values for price, \$30.69, and annual dividend, \$1.48, we find the percent yield for the stock to be

$$\text{Yld}\% = \frac{\text{Annual Dividend}}{\text{Share Price}} \times 100\% = \frac{\$1.48}{\$30.69} \times 100\% = 4.82\%$$
2. Substituting the values for price, \$62.25, and annual dividend, \$1.76, we find the percent yield for the stock to be

$$\text{Yld}\% = \frac{\text{Annual Dividend}}{\text{Share Price}} \times 100\% = \frac{\$1.76}{\$62.25} \times 100\% = 2.83\%$$

YOUR TURN 6.64

1. Find the percent yield for a stock with a price of \$37.40 and an annual dividend of \$1.60.
2. Find the percent yield for a stock with a price of \$73.22 and an annual dividend of \$2.41.

The stock table information is now, and has been, available online, from websites such as cnn.com/markets, markets.businessinsider.com/stocks, and marketwatch.com. The same information is available from these sites as from the newspaper listings, but are often accessed one stock at a time. [Figure 6.14](#) shows the stock table for Lowe's on September 7, 2022.

Lowe's Cos.

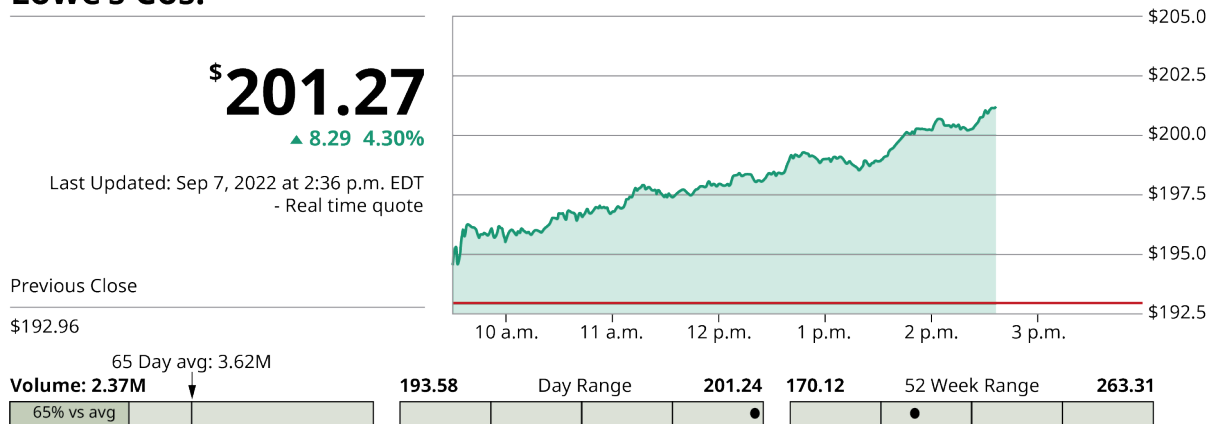


Figure 6.14 Key data for Lowe's stock 9/7/2022 (data source: marketwatch.com)

Other key data is further down on the website, and is shown in [Figure 6.15](#), below.

Key Data

Open \$193.71	Day Range 193.58–201.44
52 Week Range 170.12–263.31	Market Cap \$119.77B
Shares Outstanding 620.7M	Public Float 620.17M
Beta 1.14	Rev. Per Employee \$280.57K
P/E Ratio 15.79	EPS \$12.69
Yield 2.10%	Dividend \$1.05
Ex-Dividend Date Oct 18, 2022	Short Interest 11.09M 08/15/22
% of Float Shorted 1.79%	Average Volume 3.62M

Performance

5 Day	3.74% 
1 Month	-0.01% 
3 Month	4.65% 
YTD	-22.08% 
1 Year	-1.32% 

Figure 6.15 Key data for Lowe's stock 9/7/2022 (data source: marketwatch.com)

Notice that the 52-week high and low are now shown as the 52-week range. However, you get additional information, including the stock performance over the past 5 days, past month, past 3 months, the year to date (YTD), and over the past year. You can also read the number of shares outstanding, the expected date for the dividend (EX-DIVIDEND DATE), and importantly for the P/E ratio, the earning per share (EPS).

EXAMPLE 6.65**Reading an Online Stock Table**

Consider the stock table ([Figure 6.16](#)), and answer the questions based on the table.

McDonald's Corp.

\$258.87

▲4.48 1.76%

Last Updated: Sep 7, 2022 2:49 p.m. EDT
- Real time quotePrevious Close
\$254.39

Key Data

Open	Day Range	P/R Ratio	EPS
\$255.14	254.96–259.82	31.89	\$8.12
52 Week Range	Market Cap	Yield	Dividend
217.68–271.15	\$187.16B	2.13%	\$1.38
Shares Outstanding	Public Float	Ex-Dividend Date	Short Interest
735.72M	734.72M	Aug 31, 2022	5.53M 08/15/22
Beta	Rev. Per Employee	% of Float Shorted	Average Volume
0.76	\$117.97K	0.75%	2.47M

Performance

5 Day	2.61%
1 Month	-0.95%
3 Month	5.40%
YTD	-3.43%
1 Year	8.49%

Figure 6.16 Key data for McDonald's stock 9/7/2022 (data source: marketwatch.com)

1. What is the current price for McDonald's Corp on this date?
2. What is the 52-wk high? 52-wk low?
3. When is the dividend expected?
4. What is its yield?
5. What is the earnings per share?

✓ Solution

1. Looking at the table, the current price of a share is \$258.87.
2. The high was \$271.15, and the low was \$217.68.
3. August 31, 2022
4. 2.13%
5. The EPS value is \$8.12.

> YOUR TURN 6.65

Consider the stock table below, and answer the questions based on the table.

Intel Corp.

\$30.67

▲ 0.31 1.02%

Last Updated: Sep 7, 2022 2:54 p.m. EDT
- Real time quote

Previous Close

\$30.36

65 Day avg: 39.01M

Volume: 23.89M

61% vs avg

30.05

Day Range

30.68

30.05

52 Week Range

56.28

Key Data

Open

\$30.42

Day Range

30.05–30.68

P/E Ratio

6.57

EPS

\$4.66

52 Week Range

30.05–56.28

Market Cap

\$124.66B

Yield

4.77%

Dividend

\$0.37

Shares Outstanding

4.11B

Public Float

4.1B

Ex-Dividend Date

Aug 4, 2022

Short Interest

63.38M 08/15/22

Beta

1.21

Rev. Per Employee

\$606.06K

% of Float Shorted

1.54%

Average Volume

39.01M

Performance

5 Day

■ -3.92%

YTD

■ -40.45%

1 Month

■ -13.29%

1 Year

■ -42.75%

3 Month

■ -25.61%

Key data for Intel stock 9/7/2022 (data source: marketwatch.com)

1. What is the current price for Intel Corp on this date?
2. What is the 52-wk high? 52-wk low?
3. When is the dividend expected?
4. What is its yield?
5. What is the earnings per share?

As mentioned, stocks earn money in two ways, through dividends and increase in share price.

EXAMPLE 6.66

Dividends Paid

Darma owns 150 shares of stock in the GDW company. This quarter, GDW is paying \$0.87 per share in dividends. How much will Darma earn in dividends this quarter?

✓ Solution

Each share pays \$0.87, so Darma earns $150 \times \$0.87 = \130.50 .

 YOUR TURN 6.66

1. Yulia owns 300 shares of stock in YYZ company. It pays \$1.12 per share this quarter. How much did Yulia earn this quarter on stock in YYZ?

EXAMPLE 6.67**Stock Price Increases**

Vincent buys 100 stocks in the REM company for \$21.87 per share. One year later, he sells those 100 shares for \$29.15 per share.

1. How much money did Vincent make?
2. What was his return on investment for that one year?

 Solution

1. Vincent spent \$21.87 per share to buy the stock. The total he spent on the stock was $\$21.87 \times 100 = \$2,187.00$. When he sold the stock, the price was \$29.15, so he received $\$29.15 \times 100 = \$2,915.00$. He made $\$2,915.00 - \$2,187.00 = \$728.00$.
2. His return on investment was $\frac{\text{Earnings}}{\text{Original Price}} = \frac{\$728}{\$2,187} = 33.29\%$.

 YOUR TURN 6.67

Ginny buys 200 shares of stock in UUK company for \$9.76 per share. At the end of the year, she sells those stocks for \$10.02 per share.

1. How much money did Ginny make?
2. What was her return on investment for that one year?

 VIDEO

[Reading Stock Summary Online \(https://openstax.org/r/Reading_Stock\)](https://openstax.org/r/Reading_Stock)

Mutual Funds

A **mutual fund** is a collection of investments that are all bundled together. When you buy shares of a mutual fund, your money is pooled with the assets of other investors. This pooled money is invested in stocks, bonds, money market instruments, and other assets. Mutual funds are typically operated by professional money managers who allocate the fund's assets and attempt to produce capital gains or income for the fund's investors.

A key benefit of mutual funds is that they allow small or individual investors to invest in professionally managed portfolios of equities, bonds, and other securities. This means each shareholder participates proportionally in the gains or losses of the fund. The performance of a mutual fund is usually stated as how much the mutual fund's total value has increased or decreased. Since there are many different investments inside the mutual fund, the risk is reduced significantly, compared to direct ownership of stocks. Even so, mutual funds historically perform well and can earn more than 10% annually.

The investments that make up a mutual fund are structured and maintained to match stated investment objectives, which are specified in its **prospectus**. A prospectus is a pamphlet or brochure that provides information about the mutual fund. Before buying shares of a mutual fund, consult its prospectus, consider its goals and strategies to see if they match your goals and values and also research any associated fees.

 VIDEO

[Mutual Funds \(https://openstax.org/r/investing_basics_mutual_funds\)](https://openstax.org/r/investing_basics_mutual_funds)

EXAMPLE 6.68**Investing in a Mutual Fund**

Kaitlyn has analyzed her \$12,862.50 quarterly budget using the 50-30-20 budget philosophy, and sees she should be saving or paying down debt with \$2,572.50 per quarter. She decides to invest \$1,300 quarterly a mutual fund that reports an average return of 11.62% over the 18-year life of the mutual fund. Assuming that this interest rate continues, and is compounded quarterly, how much will her mutual fund account be worth after 5 years?

✓ Solution

Kaitlyn's plan is an ordinary annuity, and so the future value of her account can be found using the formula

$FV = pmt \times \frac{(1 + r/n)^n \times t - 1}{r/n}$, with a payment of \$1,300, a rate of 0.1162, number of compounding periods 4, after 5 years. Substituting these values into the formula and calculating, we find

$$\begin{aligned} FV &= pmt \times \frac{(1 + r/n)^n \times t - 1}{r/n} \\ &= \$1,300 \times \frac{(1.02905)^{20} - 1}{0.02905} \\ &= \$1,300 \times \frac{0.773084935178}{0.032905} \\ &= \$1,300 \times 26.6122189784 \\ &= \$34,595.88 \end{aligned}$$

Kaitlyn's mutual fund will be worth \$34,595.88 after 5 years.

> YOUR TURN 6.68

1. Aidan decides to invest \$3,200 annually in a mutual fund. He expects the fund to have a 10.8% interest rate compounded annually. How much will Aidan's mutual fund account have after 15 years?

EXAMPLE 6.69**Investing in a Mutual Fund to Reach a Goal**

Kaitlyn wants to retire with \$1,500,000 in her mutual fund account. She will invest for 35 years. The mutual fund reports an average return of 11.62% over the 18-year-long life of the mutual fund. Assuming that this interest rate continues, and is compounded quarterly, how much will she need to pay annually into her mutual fund to reach her goal?

✓ Solution

Kaitlyn's plan is an ordinary annuity, and so the payment to reach her goal can be found using the formula

$pmt = \frac{FV \times (r/n)}{(1 + r/n)^n \times t - 1}$, with a FV , or goal, of \$1,500,000, a rate of 0.1162, for 35 years. Substituting these values into the formula and calculating, we find

$$\begin{aligned} pmt &= \frac{FV \times (r/n)}{(1 + r/n)^n \times t - 1} \\ &= \frac{\$1,500,000 \times (0.1162/4)}{(1 + 0.1162/4)^{140} - 1} \\ &= \frac{\$174,300}{26.0558103113} \\ &= \$6,689.49 \end{aligned}$$

Kaitlyn needs to invest \$6,689.49 per year (or \$557.46 per month) into the mutual fund to reach \$1,500,000 in 35 years.

> YOUR TURN 6.69

1. How much does Aidan need to invest annually in his mutual fund to reach a goal of \$1,000,000 in 40 years. He expects the fund to have a 10.8% interest rate compounded annually.

Return on Investment

As in [Methods of Savings](#), the formula for return on investment is $\text{ROI} = \frac{FV - P}{P}$. As indicated before, this formula does not take into account how long the investment took to reach its current value. It depends only on the initial value, P , and the value at the end of the investment, FV .

EXAMPLE 6.70

Return on Investment for a Bond

Recall [Example 6.62](#), in which Muriel purchased a \$3,000 bond with a maturity of 4 years at a fixed coupon rate of 5.5% paid annually. What was Muriel's return on investment?

Solution

Each year, Muriel received \$165. She received this money four times, so earned a total of \$660. This represents $FV - P$, or just the earnings. Using that we find that the ROI is $\text{ROI} = \frac{660}{3000} = 0.22$, or 22%.

YOUR TURN 6.70

1. Maureen invests \$5,000 in a bond with a maturity date in 5 years at a fixed coupon rate of 4.75%. What is Maureen's return on investment?

As mentioned, the ROI does not address the length of time of the investment. A good way to do that is to equate the ROI to an account bearing interest that is compounded annually.

The annual return is the average annual rate, or the annual percentage yield (APY) that would result in the same amount were the interest paid once a year.

FORMULA

The formula for annual return is $\text{annual return} = \left(\frac{FV}{P}\right)^{\left(\frac{1}{t}\right)} - 1$, where t = the number of years, FV = new value, and P = starting principal.

We apply this to the previous example.

EXAMPLE 6.71

Annual Return on Investment for a Bond

Recall [Example 6.70](#), in which Muriel purchased a \$3,000 bond with a maturity of 4 years at a fixed coupon rate of 5.5% paid annually. What was Muriel's annual return on investment? Interpret this as compound interest.

Solution

Muriel earned a total of \$660. This represents $FV - P$, or just the earnings. The starting principal was \$3,000. The value at the end of 4 years was $\$3,000 + \$660 = \$3,660$. The time of the investment was 4 years. Using that we find that the annual return is $\text{annual return} = \left(\frac{FV}{P}\right)^{\left(\frac{1}{t}\right)} - 1 = \left(\frac{\$3,660}{\$3,000}\right)^{\left(\frac{1}{4}\right)} - 1 = 1.22^{\frac{1}{4}} - 1 = 1.050969 = 0.050969$, or 5.10%. The 5.5% bond earned the equivalent of 5.10% compounded annually.

YOUR TURN 6.71

1. Maureen invests \$5,000 in a bond with a maturity date in 5 years at a fixed coupon rate of 4.75%. What is Maureen's annual return on investment?

In [Example 6.71](#) and Your Turn, the annual return was lower than the interest rate of the investment. This is because the

interest from a bond is simple interest, but annual yield equates to compounded annually.

EXAMPLE 6.72

Return on Investment for Stock in Company ABC

Haniah buys stock in the ABC company, investing a total of \$13,000. After 20 years, the stock is worth \$132,293.49, including reinvestment of dividends.

1. What is Haniah's return on investment?
2. What is Haniah's annual return?

Solution

1. To calculate Haniah's return on investment, substitute \$13,000 for P and \$132,293.49 for FV in the formula $ROI = \frac{FV-P}{P}$ and calculate. Doing so we find Haniah's return on investment to be

$$ROI = \frac{FV-P}{P} = \frac{\$132,293.49 - \$13,000}{\$13,000} = 9.176422, \text{ or } 917.64\%$$

2. To calculate Haniah's annual return, substitute \$13,000 for P and \$132,293.49 for FV in the formula

$$\text{annual return} = \left(\frac{FV}{P} \right)^{\left(\frac{1}{t} \right)} - 1 \text{ and calculate. Doing so we find her annual return to be}$$

$$\text{annual return} = \left(\frac{FV}{P} \right)^{\left(\frac{1}{t} \right)} - 1 = \left(\frac{\$132,293.49}{\$13,000} \right)^{\left(\frac{1}{20} \right)} - 1 = 0.1230, \text{ or } 12.3\%$$

YOUR TURN 6.72

Rixie deposits \$23,000 in the stock of DEF company. After 12 years, her stock is worth \$108,501.30.

1. What is Rixie's return on investment?
2. What is Rixie's annual return?

You should see that the annual return is equal to the annual compounded interest that was assumed for the stocks.

Compute Payment to Reach a Financial Goal

As in [Methods of Savings](#), determining the payment necessary to reach a financial goal uses the payment formula for an ordinary annuity, $pmt = \frac{FV \times (r/n)}{(1 + r/n)^n \times t - 1}$. If dealing with mutual funds or stocks, an assumed annual interest rate, compounded, will be used. This value is often determined through research and informed speculation.

EXAMPLE 6.73

Richard is saving for new siding for his home. He and his partner believe they will need \$37,500 in 10 years to pay for the siding. How much should they invest yearly in a mutual fund they believe will have an annual interest rate of 12%, compounded annually, in order to reach their goal?

Solution

The necessary annual payment is found using the function $pmt = \frac{FV \times (r/n)}{(1 + r/n)^n \times t - 1}$ with $FV = 37,500$, $r = 0.12$, and $n = 1$. Substituting and calculating, we find the annual payment should be

$$\begin{aligned} pmt &= \frac{FV \times (r/n)}{(1 + r/n)^n \times t - 1} \\ &= \frac{\$37,500 \times (0.12/1)}{(1 + 0.12/1)^1 \times 10 - 1} \\ &= \frac{\$4,500}{(1.12)^{10} - 1} \\ &= \frac{\$4,500}{2.10584820834} \\ &= \$2,136.91 \end{aligned}$$

> YOUR TURN 6.73

1. Pete and Erin want to save for their child's college. They think they will need \$90,000 in 14 years. How much should they invest annually in a mutual fund they believe will yield 9.5% compounded annually?

Retirement Savings Plans

We close this section by investigating the three main forms of retirement savings accounts: traditional individual retirement accounts (IRAs), Roth IRAs, and 401(k) accounts. Each has distinct characteristics that are suited to different investors' needs.

Individual Retirement Accounts

A **traditional IRA** lets you contribute up to an amount set by the government, which may change from year to year. For example, the maximum contribution for 2022 is \$6,000; \$7,000 over age 50. Anyone is eligible to contribute to a traditional IRA, regardless of your income level. Your money grows tax-deferred, but withdrawals after age 59½ are taxed at current rates. Traditional IRAs also allow you to use the contribution itself as a deduction on a current year tax return.

Roth IRAs allow contributions at the same levels as traditional IRAs, with a maximum \$6,000 for 2022; \$7,000 over age 50. However, to be eligible to make contributions, your earned income must be below a certain level. A Roth IRA allows after-tax contributions. In other words, the contribution itself is not tax-deductible, as it is with the traditional IRA. However, your money grows tax-free. If you make no withdrawals until you are age 59½, there are no penalties. IRAs pay a modest interest rate.

In either case, IRA deposits have to be from earned income, which in effect means if your earned income is over \$6,000 (\$7,000) then you can deposit the maximum.

EXAMPLE 6.74

Comparing Roth IRAs to Traditional IRAs

Which type of IRA, Roth or traditional, has an income limit for its use?

Solution

Roth IRAs require income to be below a certain limit.

> YOUR TURN 6.74

1. Which type of IRA, Roth or traditional, allow deposits before tax, but have earnings that are taxed after the age of 59 ½?

WHO KNEW?

In 2022, the maximum that can be added to a Roth IRA was \$6,000 for those under 50 years of age. For those over 50 years of age, the maximum that can be added to a Roth IRA is \$7,000. However, to qualify for a Roth IRA in fall of 2022, a single person's modified adjusted gross income (MAGI) must be below \$129,000. Then, if a single person's income is between \$129,000 and \$144,000, the maximum contribution is reduced from the limit for incomes below \$129,000. For a married couples filing a joint tax return those values are \$204,000 to \$214,000.

401(k) Accounts

Your employer may offer a retirement account to you. These are often in the form of a **401(k)** account. There are traditional and Roth 401(k) accounts, which differ in how they are taxed, much as with other IRAs. In the traditional 401(k) plans, the money is deposited before tax is assessed, which means you do not pay taxes on this money. However, that means when money is withdrawn, it is taxed. These accounts are similar to mutual funds, in that the money is invested in a wide range of assets, spreading the risk.

One of the perks some employers offer is to match some amount of your contributions to the 401(k) plan. For instance,

they may match your deposits up to 5% of your income. This is an instant 100% return on the money that was matched.

 VIDEO

[401\(k\) Accounts \(https://openstax.org/r/401\(k\)s\)](https://openstax.org/r/401(k)s)

EXAMPLE 6.75

Matching 401(k) Deposit

Alice signs up for her employer-based 401(k). The employer matches any 401(k) contribution up to 6% of the employee salary. Alice's annual salary is \$51,600.

1. What is the most money that Alice can deposit that will be fully matched by the company?
2. How much total will be deposited into Alice's account if she deposits the full 6%?
3. How much return does Alice earn if she deposits exactly 6% in her 401(k)?

 Solution

1. The employer will match up to 6% of any employee's salary. 6% of Alice's salary is $0.06 \times \$51,600 = \$3,096$. So Alice can deposit up to \$3,096 and receive that amount in matching funds in her account.
2. Alice's contribution plus the company's contribution is $\$3,096 + \$3,096 = \$6,192$, which is the total that is deposited into Alice's account.
3. She earns a 100% return on the day she deposits her \$3,096.

YOUR TURN 6.75

Jameis signs up for his employer-based 401(k). The employer matches any 401(k) contribution up to 7.5% of the employee salary. Jameis' annual salary is \$72,800.

1. What is the most money that Jamie can deposit that will be fully matched by the company?
2. How much total will be deposited into Jameis' account if he deposits the full 7.5%?
3. How much return does Jameis earn if he deposits exactly 7.5% in her 401(k)?

401(k) plans with matching funds provide great value, as their rates of return are high compared to savings accounts, and are less risky than stocks since such funds invest across many investment vehicles. The next example demonstrates the power of constant deposits into a 401(k) plan that has some employer match.

EXAMPLE 6.76

Constant Deposits into a 401(k) Plan

Dejean begins depositing \$300 per month from his paycheck each month in his employer-based 401(k) account. The employer matches this deposit as it falls below their matching threshold. Dejean expects the return to average 10% per year, compounded annually.

1. How much will Dejean's account be worth if he keeps making those payments for 30 years?
2. What will his account be worth without the matching funds?

 Solution

1. This is a form of an ordinary annuity, so the formula $FV = pmt \times \frac{(1 + r/n)^n \times t - 1}{r/n}$ will be used. The company matches Dejean's full deposit, so each month \$600 will be deposited. He is assuming the money will compound annually, so the amount deposited each year is needed as the value of pmt. For the year, he will deposit $12 \times \$600 = \$7,200$. The rate is 0.1, the number of compounding periods is 1, and the number of years is 30. Substituting and calculating, the value of Dejean's account after 30 years will be

$$\begin{aligned}
 FV &= pmt \times \frac{(1 + r/n)^n \times t - 1}{r/n} \\
 &= \$7,200 \times \frac{(1 + 0.1/1)^1 \times 30 - 1}{0.1/1} \\
 &= \$7,200 \times \frac{(1.1)^{30} - 1}{0.1} \\
 &= \$7,200 \times 164.494022689 \\
 &= \$1,184,356.96
 \end{aligned}$$

2. This is a form of an ordinary annuity, so the formula $FV = pmt \times \frac{(1 + r/n)^n \times t - 1}{r/n}$ will be used but the deposit is now only \$300 per month without the matching funds. For the year, he will deposit $12 \times \$300 = \$3,600$. The rate is 0.1, the number of compounding periods is 1, and the number of years is 30. Substituting and calculating, the value of Dejean's account after 30 years will be

$$\begin{aligned}
 FV &= pmt \times \frac{(1 + r/n)^n \times t - 1}{r/n} \\
 &= \$3,600 \times \frac{(1 + 0.1/1)^1 \times 30 - 1}{0.1/1} \\
 &= \$3,600 \times \frac{(1.1)^{30} - 1}{0.1} \\
 &= \$3,600 \times 164.494022689 \\
 &= \$592,178.48
 \end{aligned}$$

> YOUR TURN 6.76

Crystal begins depositing \$450 per month from her paycheck each month in her employer-based 401(k) account. The employer matches \$350 of each deposit. Crystal expects the return to average 9% per year, compounded annually.

1. How much will Crystal's account be worth if she keeps making those payments for 25 years?
2. What will her account be worth without the matching funds?

Check Your Understanding

39. Which investment has the highest risk?
40. What is different about who can use a Roth IRA versus a traditional IRA??
41. How do mutual funds reduce risk?
42. How much in earnings does a 10-year bond with issue price \$5,000 that pays 4% interest annually is generated?
43. If an individual makes \$115,000 annually, is the person eligible for a Roth IRA?
44. An employer 401(k) matches up to 4% of income by employees. Merisol's annual salary is \$87,500. If Merisol wants to deposit \$4,400 annually in the 401(k), how much will be matched?
45. A person owns 300 shares of stock. It pays \$.38 per share this quarter. How much does the share owner earn from the stock that quarter?
46. David purchased stocks for \$3,500. The stocks paid dividends by reinvesting them in the stock. After 3 years, he sold the socks for \$4,650. What was David's annual return on that investment?
47. How much must be deposited annually in a mutual fund that is expected to bear 12.5% interest compounded annually if the account is to be worth \$750,000 after 25 years?



SECTION 6.7 EXERCISES

1. What is the maturity date for a bond?
2. What is the issue price for a bond?
3. Stock investments increase in value in what two ways?

4. Which is the least risky of stocks, bonds, mutual funds, CDs?
5. Which type of individual retirement account allows pre-tax deposits?
6. What are the limits on contributions to individual retirement accounts in 2022?
7. Of bonds, stocks, mutual funds, CDs, and money market accounts, which do not allow for withdrawal until a certain time period has passed?
8. Which type of IRA allows the account to grow tax free, provided no withdrawals are made until after the age of $59\frac{1}{2}$?
9. Which of bonds, mutual funds, and CDs are professionally managed?
10. Why do mutual funds and IRAs have relatively low risk?

For the bonds with the given properties, find a. the amount paid each year and b. the total amount earned with the bond.

11. Issue price of \$10,000 pays 3.5% annually, matures in 5 years.
12. Issue price of \$3,400 pays 2.75% annually, matures in 10 years.
13. Issue price of \$1,000 pays 2.8% annually, matures in 5 years.
14. Issue price of \$5,000 pays 3.75% annually, matures in 15 years.

In the following exercises, find: a. the return on investment and b. the annual return for the bond described. Round to two decimal places.

15. Issue price of \$10,000 pays 3.5% annually, matures in 5 years.
16. Issue price of \$3,400 pays 2.75% annually, matures in 10 years.
17. Issue price of \$1,000 pays 2.8% annually, matures in 5 years.
18. Issue price of \$5,000 pays 3.75% annually, matures in 15 years.
19. 40 shares of stock are owned. The dividend per share is \$0.38 for a quarter. How much was earned in dividends on this stock this quarter?
20. 150 shares of stock are owned. The dividend per share is \$0.78 for a quarter. How much was earned in dividends on this stock this quarter?
21. 100 shares of stock are owned. The dividend per share is \$0.18 for a quarter. How much was earned in dividends on this stock this quarter?
22. 250 shares of stock are owned. The dividend per share is \$0.41 for a quarter. How much was earned in dividends on this stock this quarter?

For the following exercises, 70 shares of stock were purchased for \$31.50 per share 5 years ago. Over the 5 years, the total of all dividends earned from this stock was \$6.34 per share. The stock is sold for \$34.83.

23. How much was earned with dividends and share price increase combined?
24. What was the return on investment for these stocks?
25. What was the annual return for these stocks?

For the following exercises, 10 shares of stock were purchased for \$18.91 per share 3 years ago. Over the 3 years, the total of all dividends earned from this stock was \$3.18 per share. The stock is sold for \$22.01.

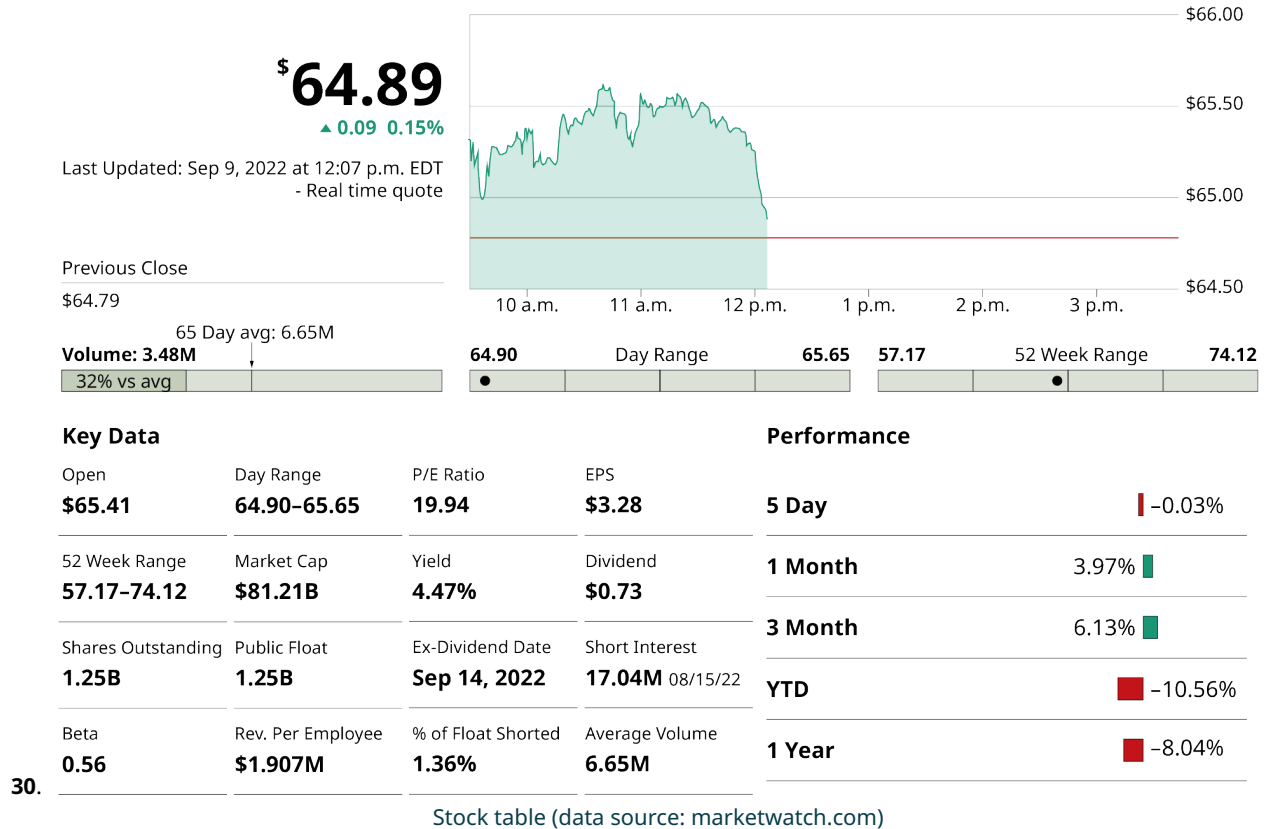
26. How much was earned with dividends and share price increase combined?
27. What was the return on investment for these stocks?
28. What was the annual return for these stocks?

For the following exercises, use the given stock table to answer the following:

- a. What was the 52-week low?
- b. What was the dividend?
- c. What is its year-to-date performance?
- d. What is its yield?



Stock table (data source: marketwatch.com)



Stock table (data source: marketwatch.com)

In the following exercises, find the future value of the mutual fund or IRA with the given annual deposit, the duration of the investment, and the assumed annual, compounded, percentage rate.

31. IRA, annual deposit = \$6,000, 25 years, assumed percentage rate of 11.3%
32. IRA, annual deposit = \$4,800, 15 years, assumed percentage rate of 9.7%

33. Mutual fund, annual deposit = \$7,500, 35 years, assumed percentage rate of 10%

34. Mutual fund, annual deposit = \$12,000, 40 years, assumed percentage rate of 9%

In the following exercises, find the annual deposit necessary, into an IRA or mutual fund, to reach the stated financial goal, given the assumed annual, compounded, interest rate, and the duration of the deposits. Convert that annual deposit to a monthly amount.

35. Mutual fund, goal = \$1,000,000, 25 years, assumed percentage rate of 11.3%

36. Mutual fund, goal = \$100,000, 15 years, assumed percentage rate of 9.7%

37. IRA, goal = \$750,000, 35 years, assumed percentage rate of 10%

38. IRA, goal = \$1,750,000, 40 years, assumed percentage rate of 9%

39. Francis's employer matches 401(k) contributions up to 7% of annual salary. She makes \$98,500. What is the maximum amount that the company will match for Francis?

40. Miles' employer matches 401(k) contributions up to 4% of annual salary. He makes \$38,500. What is the maximum amount that the company will match for Miles?

41. Ila's employer matches 401(k) contributions, up to 4% of salary. She makes \$49,000. She wants to deposit \$3,000 of her own earnings per year (\$250 per month). How much, including her employer's matching funds, will be deposited into Ila's account each year?

42. Georgia works at a company that matches 401(k) contributions up to 5.5% of salary. Georgia wants to deposit \$6,000 of her own earnings per year (\$500 per month) in her 401(k). If she makes \$65,000 annually, how much, including employer contributions, will be deposited in Georgia's account annually?

In the following exercises, Cheryl's company offers a 401(k) account to all employees. The company will match employee contributions up to 6% of the employee's salary. She earns \$81,000 per year. Cheryl decides to deposit, or contribute, \$10,000 annually in the 401(k). She expects a return of 9.5% per year.

43. How much will the company match?

44. What is her total contribution per year?

45. If she deposits into the account for 25 years, how much will her 401(k) be worth?

In the following exercises, Lavanya's company offers a 401(k) account to all employees. Her annual salary is \$58,000. The company will match employee contributions up to 7% of the employee's salary. Lavanya decides to deposit \$3,200 annually in the 401(k). She expects a return of 8.5% per year.

46. How much will the company match?

47. What is her total contribution per year?

48. If she deposits into the account for 20 years, how much will her 401(k) be worth?

In the following exercises, Ruslana wants to use her 401(k) to save \$1,400,000 when she retires in 36 years. She assumes the plan will yield 11% compounded annually. Her company will match contributions up to 5% of annual salary. Her salary is \$48,000.

49. How much total will need to be added to her account each year to reach her goal?

50. What is 5% of Ruslana's salary?

51. The company will match up to 5% of Ruslana's salary, which means half the payment (up to 5% of salary) will be contributed by the company. How much is half the necessary payment?

52. Does the answer to Exercise 51 exceed the result from Exercise 50?

53. If the answer to Exercise 52 is no, then the company contributes half the deposit in the 401(k). How much will the company contribute annually to the 401(k) if this is the case?

54. If the answer to Exercise 52 is yes, then the company contributes only the 5% match of Ruslana's salary. In this case, how much does the company contribute?

55. How much will Ruslana need to contribute per year, after the employer contribution?

56. Divide the answer to Exercise 55 by 12 to find the monthly contribution Ruslana will make.

In the following exercises, Remy wants to use his 401(k) to save \$1,750,000 when he retires in 30 years. He assumes the plan will yield 10.4% compounded annually. His company will match contributions up to 6% of annual salary. His salary is \$78,000.

57. How much total will need to be added to his account each year to reach his goal?

58. What is 6% of Remy's salary?

59. The company will match up to 6% of Remy's salary, which means half the payment (up to 6% of salary) will be contributed by the company. How much is half the necessary payment?

60. Does the answer to Exercise 59 exceed the result from Exercise 58?

61. If the answer to Exercise 60 is no, then the company contributes half the deposit in the 401(k). How much will